SSLC 2025
Chemistry

Answer Key

1. Crayolite
2. Natural Rubber
3. 6 mol
4. Oleum ($H_2S_2O_7$)
5. f
6. Alkane : C_5H_{12}, C_3H_8 Alkene : C_2H_4 Alkyne: C_7H_{12}
7. A) 64/32 = 2 mol B) 2 x 28 = 56 g
8. A) Roasting B) Calcination
9. A) $CH_3 - COOH$ B) CCl_4
10. A) $CH_3 - CH_2 OH$ or $CH_3 - O - CH_3$ B) [Ethanol or Methoxymethane]
11. A) $CuSO_4$ solution B) Prevent corrosion
    C) During electroplating of copper on iron using copper sulfate solution, the blue colour of the copper sulphate solution remains unchanged because for every copper ion deposited at the cathode, an equivalent amount of copper is oxidized at the anode, maintaining the concentration of copper ions in the solution.
12. A) (ii) The 4s level is filled before 3d level due to lower energy of 4s than 3d
    B) s – block
13. A) I B) (i) Decreases (ii) Increases
14. A) (i) Addition (ii) Polymerisation
    B) Poly Vinyl Chloride Manufacture of pipes
15. A) (i) 20 (ii) ½ X
    B) 10
16. A) 25 B) +7 C) Period – 4, Group – 7
    D) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$
17. A) $BaCl_2$ B) $BaSO_4$ C) Precipitate remain unchanged
    D) Sulphate salts
18. A) 3 B) Anode – Mg Cathode – Ag
    C) Anode: $Mg \rightarrow Mg^{2+} + 2e^-$ Cathode: $Ag^+ + 1e^- \rightarrow Ag$
19. A) Heamatite – Fe_2O_3 B) $CaO + SiO_2 \rightarrow CaSiO_3$
    C) Iron pyrites D) Stainless steel and Nichrome